Robótica de Manipulação: 7^{th} Problems Class

André Carvalho

Note Title

2021-04-22

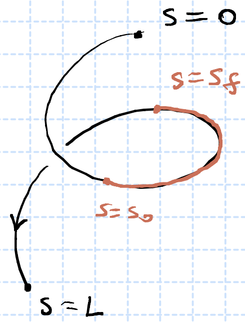
1. (a)

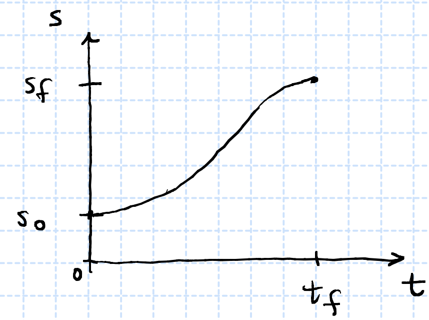
$s \leftarrow$ length

$s_0 \leftarrow$ initial length

$s_f \leftarrow$ final length

$t_f \leftarrow$ interval of time





parameterize the length as a function of time, $s = s(t)$.

Cubic polynomial, $s(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$. (1)

$$\text{Constraints} \begin{cases} s(0) = s_0 , & (2) \\ s(t_f) = s_f , & (3) \\ \dot{s}(0) = 0 , & (4) \\ \dot{s}(t_f) = 0 . & (5) \end{cases}$$

speed, $\dot{s}(t) = a_1 + 2a_2 t + 3a_3 t^2$. (6)

$t = 0$, $(1) \Rightarrow s(0) = a_0 \overset{(2)}{\Rightarrow} a_0 = s_0$, $\quad(6) \Rightarrow \dot{s}(0) = a_1 \overset{(4)}{\Rightarrow} a_1 = 0$. $\Rightarrow \begin{cases} s(t) = s_0 + a_2 t^2 + a_3 t^3 & (7) \\ \dot{s}(t) = 2a_2 t + 3a_3 t^2 & (8) \end{cases}$

$t = t_f$, $(7) \Rightarrow s(t_f) = s_0 + a_2 t_f^2 + a_3 t_f^3 \overset{(3)}{=} s_f$, $\quad(8) \Rightarrow \dot{s}(t_f) = 2a_2 t_f + 3a_3 t_f^2 \overset{(5)}{=} 0$ $\Rightarrow \begin{cases} t_f^2 a_2 + t_f^3 a_3 = s_f - s_0 \\ 2t_f a_2 + 3t_f^2 a_3 = 0 \end{cases}$

$$\begin{pmatrix} t_f^2 & t_f^3 \\ 2t_f & 3t_f^2 \end{pmatrix} \begin{pmatrix} a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} s_f - s_0 \\ 0 \end{pmatrix} \Leftrightarrow A x = b \Rightarrow x = A^{-1} b$$

$$\boxed{\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}$$

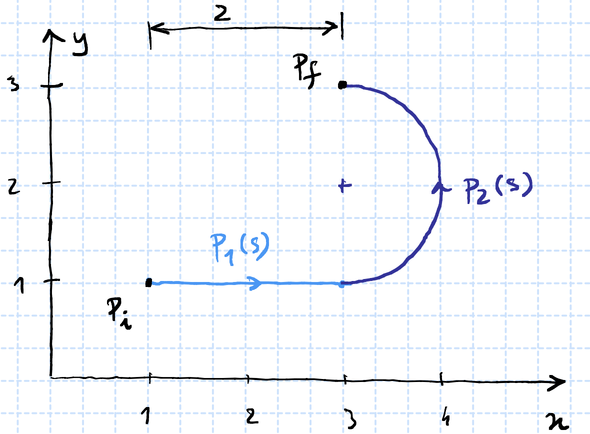
$$\begin{pmatrix} a_2 \\ a_3 \end{pmatrix} = \frac{1}{\underbrace{3t_f^4 - 2t_f^4}_{= 1/t_f^4}} \begin{pmatrix} 3t_f^2 & * \\ -2t_f & * \end{pmatrix} \begin{pmatrix} s_f - s_0 \\ 0 \end{pmatrix}$$

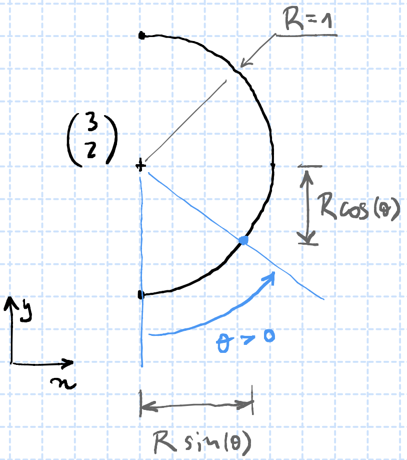
$$\Rightarrow \begin{pmatrix} a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} \frac{3}{t_f^2} & * \\ -\frac{2}{t_f^3} & * \end{pmatrix} \begin{pmatrix} s_f - s_0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{t_f^2}(s_f - s_0) \\ -\frac{2}{t_f^3}(s_f - s_0) \end{pmatrix}$$

$s(t) = s_0 + a_2 t^2 + a_3 t^3$, $\quad s(t) = s_0 + \frac{3}{t_f^2}(s_f - s_0) t^2 - \frac{2}{t_f^3}(s_f - s_0) t^3$.

3. (a)

$$p_1(s) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} s \\ 0 \end{pmatrix}, \quad 0 \le s \le 2$$

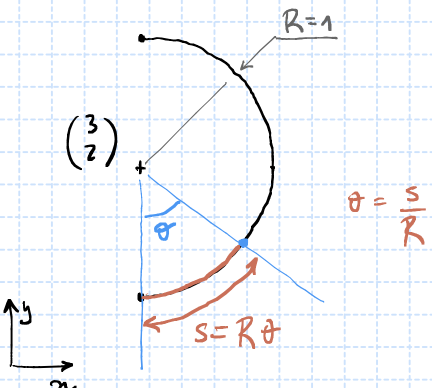




$$p_2(\theta) = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} R\sin(\theta) \\ -R\cos(\theta) \end{pmatrix}, \quad 0 \le \theta \le \pi$$

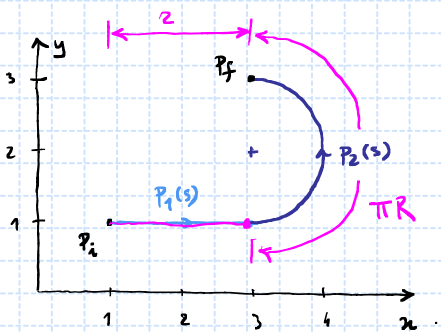
$$p_2(s) = \begin{pmatrix} 3 + R\sin\left(\frac{s}{R}\right) \\ 2 - R\cos\left(\frac{s}{R}\right) \end{pmatrix}, \quad 0 \le s \le \pi R$$

$$0 \le \frac{s}{R} \le \pi$$



Gathering,

$$p(s) = \begin{cases} p_1(s), & 0 \le s \le 2 \\ p_2(s-2), & 0 \le s-2 \le \pi R \end{cases}$$



$$p(s) = \begin{cases} \begin{pmatrix} s+1 \\ 1 \end{pmatrix}, & 0 \le s \le 2 \\ \begin{pmatrix} 3 + R\sin\left(\frac{s-2}{R}\right) \\ 2 - R\cos\left(\frac{s-2}{R}\right) \end{pmatrix}, & 2 \le s \le 2+\pi R \end{cases}$$

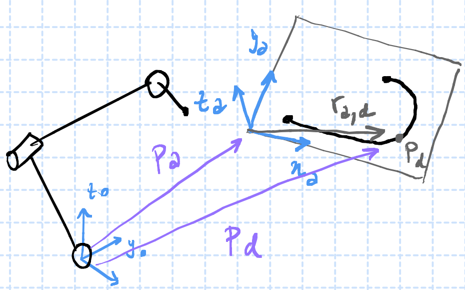
$R = 1$

$L = 2 + \pi R$

(b)

$$s(t) = \left[s_0 + \frac{3}{t_f^2}(s_f - s_0)t^2 - \frac{2}{t_f^3}(s_f - s_0)t^3 \right] \Bigg|_{\substack{s_0 = 0 \\ s_f = 2+\pi}}$$

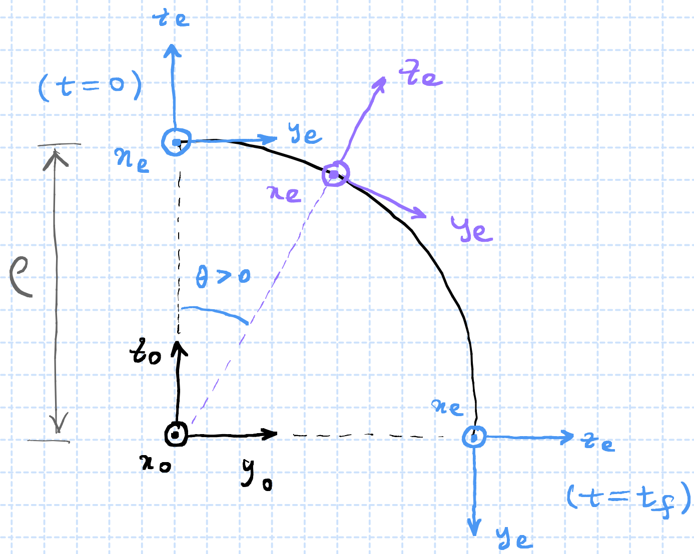
(c)



$$p_d^0 = p_a^0 + R_a^0 r_{a,d}^a$$

4.

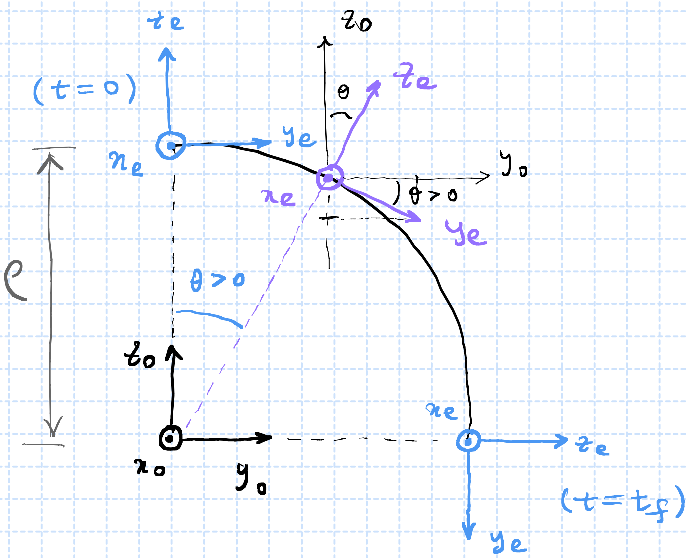
$$P_d(\theta) = \begin{pmatrix} 0 \\ \rho \sin(\theta) \\ \rho \cos(\theta) \end{pmatrix}, \quad 0 \leq \theta \leq \frac{\pi}{2}$$



$$s = \theta \rho \quad , \quad \theta = \frac{s}{\rho}$$

$$P_d(s) = \begin{pmatrix} 0 \\ \rho \sin\left(\frac{s}{\rho}\right) \\ \rho \cos\left(\frac{s}{\rho}\right) \end{pmatrix}, \quad 0 \leq s \leq \frac{\pi \rho}{2}$$

$$0 \leq \frac{s}{\rho} \leq \frac{\pi}{2}$$



$$R_d(\theta) = \left(x_e^0 \; y_e^0 \; z_e^0 \right)_d (\theta)$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\theta & s_\theta \\ 0 & -s_\theta & c_\theta \end{pmatrix}, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

$$R_d(s) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(s/\rho) & \sin(s/\rho) \\ 0 & -\sin(s/\rho) & \cos(s/\rho) \end{pmatrix}, \quad 0 \leq s \leq \frac{\pi \rho}{2}.$$